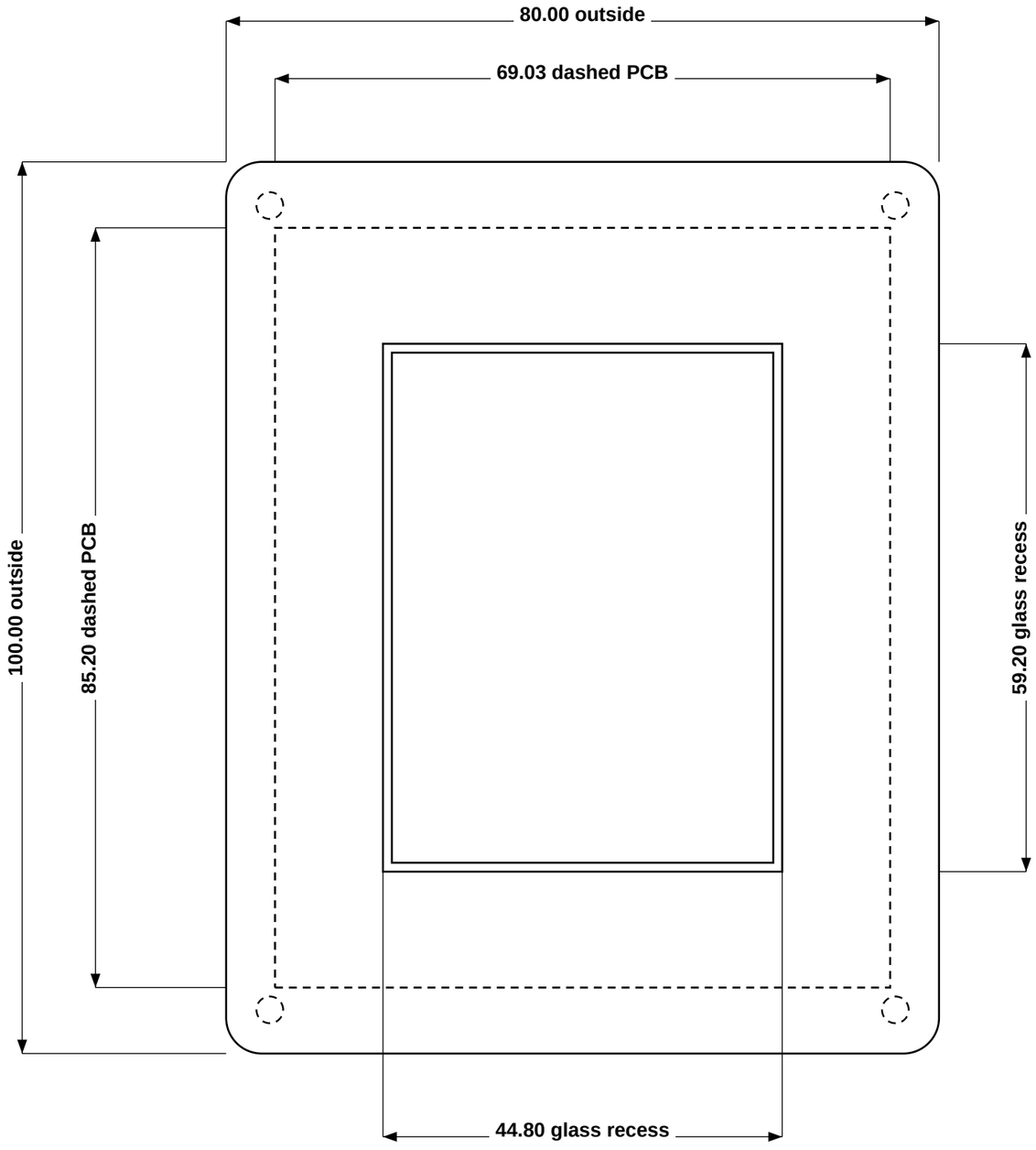
TOP COVER - front view

Outer outline, screen openings, and projected display PCB



TOP DETAILS

Outer corner radius: R4

Front and wall nominal thickness: 2.50

Top cover overall height: 35.00

Glass recess: 44.80 x 59.20 x 1.50 deep

Through-window: 42.80 x 57.20

Both openings are centered.

Dashed rectangle: assumed display PCB

69.03 x 85.20, centered.

Dashed circles: projected blind M3 holes.

PCB pocket: 69.43 x 85.60

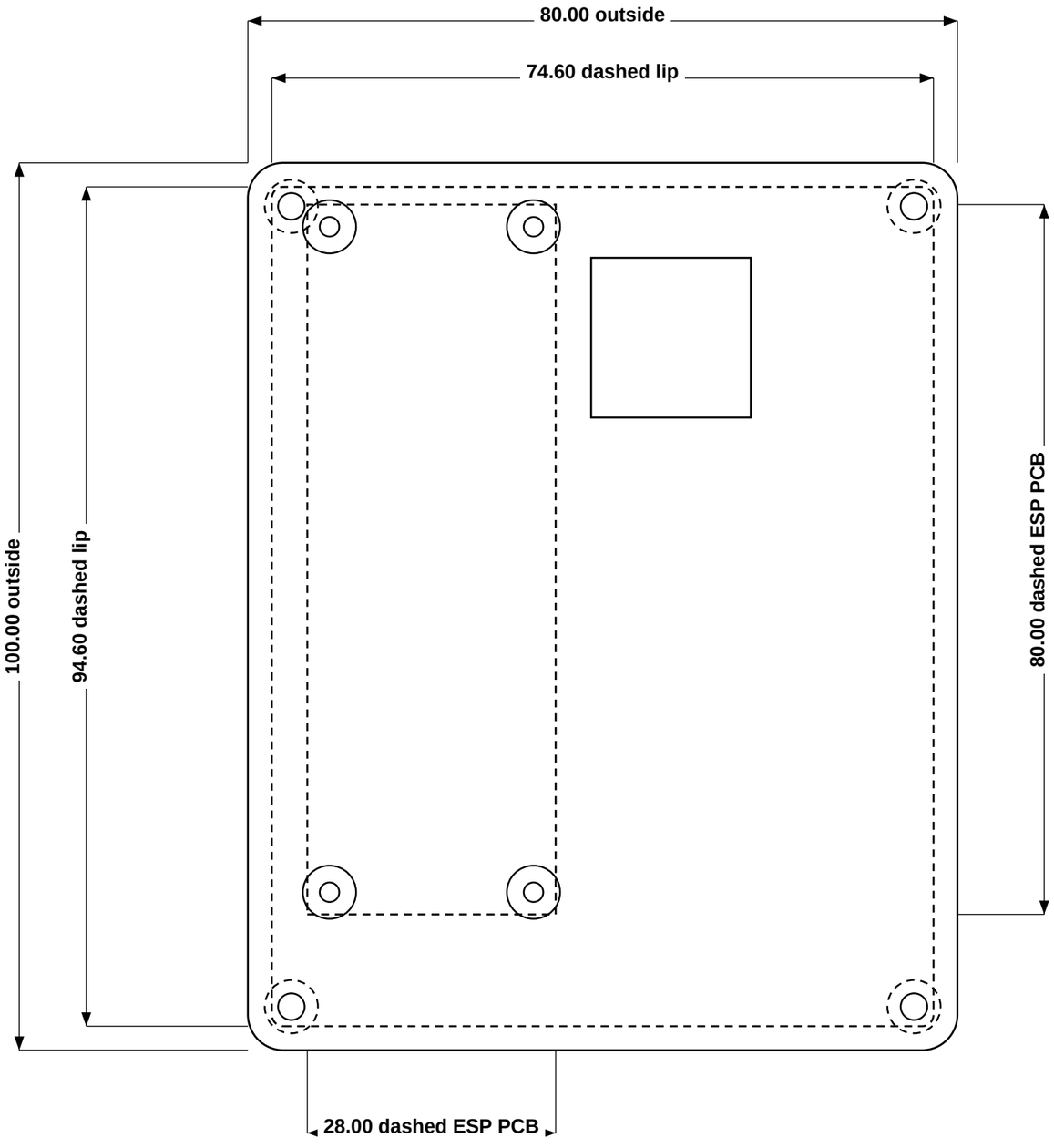
Cut from z = 1.50 to 2.60.

Pocket is not drawn as a separate outline.

See sheet 3 for important fit warnings.

BOTTOM PLATE - inside view

Dashed outlines show the lip outer boundary and ESP32 PCB



BOTTOM DETAILS

Plate: 80.00 x 100.00 x 2.50

Lip: 2.00 high; nominal wall 1.50

Lip outside R1.30; clearance to wall 0.20

Dashed ESP PCB: 28.00 x 80.00

Board origin: X 6.70, Y 15.30

4 standoff centres: X 9.20 / 32.20;

Y 17.80 / 92.80 (pattern 23 x 75).

Standoffs: Ø6 x 4 high, hole Ø2.2.

Corner holes: Ø3 through, Ø6 counterbore

2.00 deep from bottom; 4 locations.

Counterbores shown dashed (opposite side).

Centres: X 4.90 / 75.10;

Y 4.90 / 95.10 (pattern 70.20 x 90.20).

Knockout: 18.00 x 18.00;

origin X 38.70, Y 71.30; 0.50 skin.

Coordinates from lower-left outside corner.

FIT CHECK AND CRITICAL DIMENSIONS

Calculated from the supplied SCAD, not a physical fit test

1. Newhaven display / breakout PCB

SCAD assumes a centered PCB outline of 69.03 x 85.20. No independent drawing for the breakout board was available to confirm these dimensions, its thickness, mounting holes, connectors or cable envelope.

Inside wall opening: 75.00 x 95.00. PCB fits the straight walls by 2.985 per side in X and 4.90 in Y.

At corner posts (4.80 square), the board overlaps their X ranges but clears them by only 0.10 in Y.

The 0.20-per-side PCB pocket intersects the posts by 0.10 in Y at the front. Check print tolerance.

The model has no dedicated display-PCB mounting holes or retainers. Do not rely on friction alone.

2. Front ledge / enclosure height

The glass recess ends at z = 1.50; the larger PCB pocket begins at the same z = 1.50 and removes the intended one-millimetre support ledge between the recess and the smaller through-window.

The top cover is 35.00 high and the base adds 2.50: assembled exterior height is about 37.50.

The 2.00-high base lip fits inside the top opening; it does not add to assembled exterior height.

3. Before printing or ordering

Measure the actual display breakout PCB and share its drawing or exact part number to confirm fit.

Check the display glass and connector depths, screw engagement, board support, and 3D-printer clearance.

These sheets document the current model only; they do not modify or repair the original SCAD.